INSTRUMENT CLUSTER (Continued)

ACTUATOR TEST

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

The instrument cluster actuator test will put the instrument cluster into its self-diagnostic mode. In this mode the instrument cluster can perform a self-diagnostic test that will confirm that the instrument cluster circuitry, the gauges, the PCI data bus message controlled indicators, and the chime tone generator are capable of operating as designed. During the actuator test the instrument cluster circuitry will sound the chime tone generator, position each of the gauge needles at various calibration points, illuminate all of the segments in the Vacuum-Fluorescent Display (VFD) unit, turn all of the PCI data bus message-controlled indicators on and off again, and display messages in the VFD for certain faults that have been set. It is suggested that a note pad and pencil be used to write down any fault information that is displayed during the test for reference.

Successful completion of the actuator test will confirm that the instrument cluster is operational. However, there may still be a problem with the PCI data bus, the Powertrain Control Module (PCM), the Airbag Control Module (ACM), the Sentry Key Immobilizer Module (SKIM), or the inputs to one of these electronic control modules. Use a DRBIII® scan tool to diagnose these components. Refer to the appropriate diagnostic information.

If an individual gauge does not respond properly, or does not respond at all during the actuator test, the instrument cluster should be removed. However, check that the four screws securing the inoperative gauge to the instrument cluster electronic circuit board are properly tightened before considering instrument cluster replacement. If the gauge mounting screws check OK, replace the faulty instrument cluster.

(1) Begin the test with the ignition switch in the Off position.

(2) Depress the odometer/trip odometer switch button.

(3) While still holding the odometer/trip odometer switch button depressed, turn the ignition switch to the On position, but do not start the engine.

(4) Release the odometer/trip odometer switch button.

(5) The instrument cluster will automatically begin the actuator test sequence, as follows:

(a) The cluster will generate a single chime tone to confirm the functionality of the chime tone generator and the chime control circuitry.

(b) The cluster will scroll the number “8” across the odometer/trip odometer VFD to confirm the functionality of all VFD segments and their control circuitry.

(c) The cluster will illuminate the decimal point in the odometer/trip odometer VFD to confirm the functionality of this VFD segment and its control circuitry.

(d) The cluster will display the EMIC software level in the odometer/trip odometer VFD (example: “SOF 8.9”).

(e) The cluster will display the last six digits (sequence number) of the Vehicle Identification Number (VIN) in the odometer/trip odometer VFD.

(f) If any faults have been set by the cluster, the cluster will display the fault information in the odometer/trip odometer VFD INSTRUMENT CLUSTER FAILURE MESSAGE. If no faults have been set, the cluster will scroll “no FAULtS” across the odometer/trip odometer VFD.

(g) The cluster will turn on, then off again each of the following indicators, one at a time, in sequence to confirm the functionality of the indicator and the cluster control circuitry:

- High Beam
- Brake
- Seatbelt
- MIL
- Check Gauges
- Low Fuel
- 4WD
- SKIS
- Cruise

(h) The cluster will sweep the needles for each of the following gauges, one at a time, to several calibration points in sequence to confirm the functionality of the gauge and the cluster control circuitry:

- Speedometer
- Fuel
- Temperature
- Tachometer
- Voltage
- Oil Pressure